Things You Thought You Knew – Your Zodiac is Wrong

Podcast: Startalk Podcast

AI Executive Summary

Core Thesis The episode serves as a scientific debunking of several widely held misconceptions regarding astrophysics and celestial mechanics. The central argument is that popular "common knowledge"—specifically concerning Einstein’s theories of relativity, the mechanics of gravity assists in space travel, and the prominence of the North Star—is often scientifically inaccurate, simplified, or outdated due to Earth's axial precession.

Key Arguments & Data

- **Einsteinian Relativity and Time Dilation:**
    - **Special Theory of Relativity:** As an object's speed increases relative to an observer, time slows down for that object (time dilation).
    - **General Theory of Relativity:** Increased gravitational field strength also causes time to slow down.
    - **GPS Utility:** Because GPS satellites are in a weaker gravitational field than observers on Earth, their clocks tick faster; engineers must "pre-correct" these signals to ensure terrestrial accuracy.
    - **Experimental Evidence:** High-speed particles in accelerators (such as muons) demonstrate longer decay times, providing empirical proof that time dilates at high velocities.
    - **The Photon Perspective:** For light traveling at the speed of light, time does not exist; therefore, from a photon's perspective, its emission and arrival are instantaneous, regardless of distance.
- **Mechanics of Gravity Assists:**
    - **"Cosmic Drafting":** A gravity assist is not merely a "slingshot" caused by gravity pulling a craft in; rather, it is the process of a spacecraft "stealing" orbital energy from a planet.
    - **The Vector of Approach:** To gain speed, a spacecraft must approach a planet from *behind* its direction of orbital motion.

Conversely, approaching from the front can act as a "brake" to slow the craft down.
    - **Mission Application:** The Voyager 1 and 2 probes utilized multiple planetary assists to accumulate enough energy to exit the solar system without requiring massive fuel expenditures.

- **The Reality of Polaris (The North Star):**
    - **Brightness Misconception:** Contrary to popular belief, Polaris is not among the brightest stars; it is ranked approximately 49th in brightness in the night sky. Most "bright stars" people mistake for the North Star are actually planets like Venus.
    - **Axial Precession:** The Earth's axis wobbles on a ~26,000-year cycle. This means the "North Star" is temporary; in roughly 14,000 years, the star Vega will become the new pole star.
    - **Navigational Accuracy:** Polaris does not sit exactly above the North Pole; it traces a small circle in the sky due to this wobble.
    - **Astrological Discrepancy:** Because of precession, the sun is no longer in the zodiac constellations assigned to specific months 2,200 years ago. The positions have shifted significantly (e.g., an astrological Libra may actually be in Scorpio).

Actionable Takeaways

- **Verify Astronomical Observations:** When identifying objects in the night sky, do not assume the brightest object is a star; check if it is a planet (such as Venus), which can mimic stellar brightness.
- **Apply Critical Thinking to "Lore":** Recognize that cultural myths (like "wishing on a star" or the permanence of the North Star) are often based on outdated observations and do not reflect current scientific reality.
- **Use Navigational Heuristics:** For basic orientation in the Northern Hemisphere, use the "lip" of the Big Dipper as a pointer toward Polaris to find the approximate direction of the North Pole.

Full Transcript

Hello, Star Talkians. Coming up, we've got another Things You Thought You Knew episode. This time, Chuck and I riff on Einsteinian time dilation, the mysterious gravity assists that space probes get, and the North Star. Spoiler alert, it's not all it's cracked up to be. Coming up on Star Talk. Welcome to Star Talk, your place in the universe where science and pop culture Collide. Star talk begins right now. I still lose sleep over this fact. OK. I still lose sleep over it. And I want you to lose sleep, too. I don't want to be the only one staring at the ceiling at night. I've got to tell you, a lot of things already beat you to that. I mean, the asteroid strike that could happen or the global warming. So here it is. You may remember or you may have heard that Einstein's relativity, more specifically, you learn this in special theory of relativity, where the faster you move, the slower time ticks for you, as others view it. Relative to the observer. Relative to the observer, correct. You don't know anything's happening. Your clock still ticks as far, you still got your heartbeat, all of this.

Okay. So this is not a physiological thing. It is an actual property of the fabric of space and time under those conditions. Wow. Okay. That is fascinating. It is. It's completely fascinating. It's completely. So I watch you fly by and the faster you go, the slower time ticks for you. Okay. But my time stays the same. To you. To me. Right. To you. So not only does speed do this, also the strength of a gravitational field will have the same effect on you. The stronger the gravitational field is, the slower time ticks for you. And that wasn't formulated until his general theory of relativity 10 years later. That took a lot more math and deeper insights.

into the universe. So once again, it makes sense because you're dealing with the fabric of space-time. It's a fabric of space-time. Correct. Wow. Okay. But initially, when formulated, you're thinking it's just because you're moving. Right. But it's actually way deeper than that. And that's why that one... Oh, by the way, his original special theory of relativity was not called that. The title of the research paper was on the electrodynamics of moving bodies. What? OK, that was the title. We would later call it the special theory of relativity, because it was a special case of what would later then

be called the general theory of relativity. OK, so the geosynchronous satellites are like middle orbit. That is far enough away from Earth's source of gravity for them to have a difference, a measurably different space time condition. Oh. So that their clocks tick faster than our clocks on Earth's surface.

because they're farther away. Remember I said, the more intense the gravitational field, the slower the time takes. So they're farther away, their time takes faster relative to us. But we get precise timings from geosynchronous satellites. So how does this work? We pre-correct the time signal from the GPS satellites to compensate for Einstein's general theory of relativity so that by the time the time reaches us, where it's been properly corrected and it matters to us and our space time continuum, not the one that's in Middle Earth orbit. Oh my goodness. So GPS couldn't work as accurately as it does. However, wait a minute. I'm sorry, just for me, you would have to have a standard in order to pre-correct something.

You know the rate at which its time is speeding up, because you can calculate what the gravitational field is up there. There you go. There you go. Okay. Okay. And once you calculate that, then you do when you say, oh my gosh, the formula works, Einstein was right. Right. This is not just something on high theory, hypothesize that may or may not be true. It is true. I just want to be there when that phrase is uttered. Oh my God, Einstein was right. Yeah, these aren't just cult heroes that we wish they were right. Right. These are real. This is real. OK, this is the real universe we're talking about here. OK, so now watch. Let's keep going faster and faster. Let's go half the speed of light, three quarters of speed, 90 percent the speed of light, 99 percent the speed of light. Time is taking slower and slower and slower.

For you you will watch the whole future history of the universe unfold in front of your eyes as Fractions of a second go by for you as you go 99. There's a formula for this of course, but 99 99.9 99.99% the speed of light because you're the observer of that of that you Your time has slowed to the point where as you observe the things that are not Moving at the speed of light, you see them... Speeding out. Basically, you see them unfold. Correct. Boom. Correct. That is insane. So now watch. Oh my God, wait. Now

watch. Okay. Wow. So, I didn't even get to the part where I lose sleep. Okay. Okay. That's what happens. By the way, by the way, there are particles that decay. You heard of like radioactivity?

Right? Yes. Right. So one part of it decays and becomes another part and releases the often deadly energy. Yes. Okay. Radioactivity. All right. Some particles. Let's take, for example, the proton or the muon. One of these decays in like six minutes when it's left out in the wild. Okay. When it's not part of an atom. Okay. When it's not in captivity. I forgot which of these. It doesn't matter for my example. Decays in like six minutes. Okay. Okay. What happens if you take that particle and speed it up in a particle accelerator? So you take a community of these particles speed them up Calculate wait a minute the internal wristwatch on these clocks says they should live longer and sure enough Their decay time takes longer Wow That's oh my god. Yes. Yes. Oh wow. Yes, and that's because That becomes living proof of what Einstein said. Yes, because we can't go half the speed of light. Right. So you can accelerate a particle to do that. Yes. And it has an internal built-in clock that decays after a certain amount of time. And there they are taking longer to decay in the exact amount that Einstein predicts. That is genius. So we don't make this stuff up, OK? Yes, it is genius. It's like triple genius. All right. Now, let's take this to an extreme level.

Let's go so fast that we're going the speed of light itself. Right? Well, we can't do that because we're made of material substance. And there's no way to do that. But there are things that travel at the speed of light. And what is that? Those would be photons. Yes. Thank you. Those would be following your finger gesture. That would be, I would say, what travels at the speed of light, light does. Yeah. I was about to say light. So if photons had a clock, the clock would never tick. Nice. Which means when I go to a mountaintop, as I did in graduate school, to observe the center of the Milky Way galaxy for stars that emitted their light 30,000 years ago, traveling through the gaps of interstellar space, and they move through space, come through Earth's atmosphere, they come down into the telescope, they reflect, go back to my detector and land on my CCD chip. That photon, when it was born, at the star that emitted it, was detected at my telescope in the same instant,

according to the photon itself. Live fast, die hard. Photon has no knowledge of that trip because time did not exist for it.

Oh, that's great. And I'm saddened that many of these photons like hit people's tanning on the beach, you know. And imagine traveling 30,000 years and land on someone's buttocks. Very sad. When the telescope I collected, I'm now decoding the nature of the universe. And some photons go right on by Earth and are still moving, but they have no internal time clock. Wow. And they're just traveling through like, I hope when I'm born, I hit a telescope. Bam! You're born on your own telescopes. Yeah, yeah, that would be interesting, but what do I want to be when I grow up? Or the instant I am emitted, where am I going to be? Where would I like to be? Right, right, right. And so just the idea that light can move across the universe and have no, and not age, this is a fascinating fact to me.

Wow, yeah, that is that is more than fascinating. Yeah. Yeah. I mean, that's mind boggling. Yeah. Yeah. So that is light the age of light. It's like it doesn't age at all. I am convinced that all of you guys secretly microdose and smoke weed. I'm just another episode. That's what I'm talking about. No. I'm Alikon Hemraj and I support StarTalk on Patreon. This is StarTalk with Neil deGrasse Tyson. Do you like these, these explainer? Of course. I won.

because Neil Splaining has always been good for me. Neil Splaining, that's the thing. Neil Splaining has always been good for me. That's a good thing, okay. You know? All other Splainings are not good. Man's Splaining. Man's Splaining ain't good. You know what I mean? Black's Splaining, white's Splaining ain't good. But Neil's Splaining, it works. That's a whole other thing. Yeah. Thank you for that, those words of encouragement. So I'll just continue. Cool. And what do you got? So I got today, we've all heard of Gravity Assist. For a spacecraft? Yes, that was in Space Jam, right? It's a gravity assist. Oh, that was, oh, that's it, people. I just retired. It's over. Because that was the worst I've ever done in my life. Jesus. So a gravity assist, the way people typically think about it is you send out a spacecraft and you want it to give it more speed than it currently has. Right. And so you find some planet that it sort of falls towards, and of course, it'll speed up as it does that. And we think of it as kind of a slingshot. Obviously, it

falls not towards it, but towards the side of it. Yes. Otherwise, it's not a gravity assist. It's a gravity crash. Exactly. So you want it to sort of swing by, and it picks up speed in so doing, and it flings out the other side. And if you angle it right and position it right, you could head towards your next destination. And as a result, you can have multiple destinations in the same voyage. So famously, Voyager 1 and 2, this was back in the 1970s when it was launched, they had trajectories that, well, let me say that differently. The solar systems planets were configured in such a way that it could get multiple planetary assists, multiple gravitational assists. And by the time it was done, it had so many gravitational assists, it had enough energy to leave the solar system entirely. And you could end up doing that without using a very big rocket to begin with. So you're basically exploiting the gravitational attraction of planets for your speed rather than spending money on fuel.

So it's the poor man's way to get out of the soul system. It's basically cosmic drafting. Drafting? Oh, I like that. Cosmic drafting is just like, why should I pedal? Let this sucker pedal for me. Think about this. As you approach a planet and its gravity attracts you, Everyone has said and it believes and thinks that it pulls you in and then flings you out the other side. But wait a minute. Yeah. Just like, uh, just like, uh, there used to be from me with women at a nightclub. Oh, is that how that worked for you? There was an attraction. Pull me in. And they'd be like, Oh no, no, no, no, no. You got out the other side. See you. See your problem. Completely symmetric picture there. Right. Exactly. Um, so the problem.

with thinking of it that way is the same gravity that's pulling you in is also preventing you from leaving. Right. OK, so you speed up as you fall in. But now as you try to escape, the gravity is saying, no, I'm pulling you back. Right. So it turns out if you look at the pure gravity picture, it is exactly symmetric to the planet. Right. Okay, the planet pulls you in, you gain speed. You have maximum speed right when you're passing the planet. Now you want to exit the planet, the planet starts pulling you back. Every time I try to get out, your speed falling in and your speed going back is an exact mirror image of your emotion.

OK, so then the question is, how do I end up getting all this damn speed? Where are the slingshot effect? OK, this is something that's hardly ever explained. And that's why we have these explainer videos. Cool. So here's the answer. It turns out if you approach a planet from behind. OK. OK. There it is. We're beating the sun. Right. And you approach it from behind in its orbit. Right. Not only will the gravity pull you in, but you will gain speed just to catch up with the planet in its orbit. That makes sense. Yes. Right. How are we going to catch up? How are we going to hit the planet come near the planet unless you have the planet's orbital speed? Right.

The planet's orbital speed has nothing to do with the planet, with the planet's gravity. Correct. Okay. So even though the gravity is symmetric, all the speed and energy you gain falling in, the planet takes back away from you. That's symmetric. Right. But the speeds you get from catching up with the planet in the orbit is not symmetric. Bro, look at that. Okay. And so what actually happened is you were tugging on the planet, okay? As the planet was moving and you were speeding up because of this. But what it means is you stole some of the planet's orbital energy to do this. You don't tell you about that. That's no they don't. So you basically, you approach the planet from behind. And you just like, don't make a ghetto mugging out of this.

Give me some of that speed. Don't turn around. Don't look at my face. Don't look at my face. Just give me the speed. Nobody needs to get hurt here. And then you take it and you keep going and you got the thing and it doesn't. And it doesn't have it and you have it. So these are, this is the art of the slingshot. That's pretty amazing. Now it makes sense. Yeah. You could slingshot off of anything. Right. We've had planets orbit the sun launched from Earth and then slingshot around Earth twice. Right. Wow. OK. Now it takes longer, right? That's if you're not in a hurry, you do this. And use Earth to slingshot. You could slingshot off the moon. You could slingshot off of Mars.

Okay, it doesn't matter what planet you want a slingshot from right in the case of Voyager a Voyager one I think what it's slung shot what's past tense of to slingshot it's slung-shotted Slingshot It was slang shot. It's slang shit off of off of Jupiter. Right. And so people think Jupiter has big gravity will

get a huge speed. It's only getting Jupiter's orbital speed. Doesn't make that clear. Look at that. And Jupiter is traveling slower than planets that orbit closer in. So the gravitational pull cancels out. But the orbital speed is what's left over. And that's what you ended up stealing in order to get the speed to go. And if you want an inverse mugging, which no one will actually want to do in space, if you you can fall towards the planet, right? Opposite the direction it's traveling. Oh, wow. OK. And if you do that, then you have sped up the planet. It eats some of your or your your trajectory's energy and you'll come out the other side moving slower than you did before.

Oh, so you can actually use it as a break as a break. That's correct. Oh, wow. OK. This is that's fascinating stuff. If you want if you needed to break in space, that's a way it takes a long time to like line up and make it all happen. So there you have it. Wow. Look at that cosmic drafting and space breaking. That's amazing. And so the more the more times you do this, the more energy you have, the more speed you accumulate. And that's how we leave the solar system. Cool. Yeah. And you might ask, but you didn't. If you slingshot off of Mars like 20 times, what happens to the Mars orbit itself if you're taking away its energy? You haven't asked that. No, I don't. Because you don't care. I was going to say, I'm not worried about Mars. I think Mars is going to be OK. But anyway, what does happen, let's say?

Over, over, over, over, over. I mean, you're basically siphoning off a tiny little bit of energy. Siphoning off a tiny little bit of energy. So, okay, so here's the thing. You know, maybe in a trillion years, if you did this every day, it would matter, but the mass of these objects relative to the mass of our measly probes, that ratio is so huge that it's like a nat flying full speed ahead into an elephant. The elephant doesn't say, but you know, watch out. Quit your chevin. The mass difference is so huge that it's not relevant. It's not important in what's going on in the solar system. But in addition, take Earth, for example, in any given day, we plow through 100 tons of meteors every single day. Wow. Tons. So this stuff falling and this stuff. So it's don't worry about it.

We have other things we should be worrying about, like climate change. Yes, tell me about it. So if you were worried about, oh my gosh, how about

Mars? Where about your own damn planet? How about that? I'm all about that. Speaking of Mars, you said, worry about your own planet. Don't worry about Mars. And so I got both of them in one. I got a Mars. from the Out of This World collection from Soulguard, right? And check this out. Every single suitcase that they make pulls about 229 bottles of plastic out of the environment. So here we are with Mars, and we worried about our own planet at the same time. Wait, wait, so you're saying I should drink more plastic bottles so you can make more of that luggage? No.

I'm saying for the idiots that drink plastic bottles, this is part of the solution. And for the smart people who want to buy this and help, that's what they should do. OK, so it's not only gets the bottles out. If you like the universe, if you love the universe, you got one that looks like Mars. Yep. That's right. That's very cool. 12 million plastic bottles pulled in 2021. Right. See, that's what we're doing here with Chuck. Do I recognize Valis Marineris on that? Really? Don't pretend like... I'm like, because here's what I'm saying. Do you? No, it looks like a high resolution image of Martian surface. It's not just like red, right? No. Pretending to be Mars. It is Mars. It's got very...

good detail on the surface. So highly recognizable if you're Mars fluent, which apparently you're not. No. OK. I don't speak Martian. I'm sorry. Valos Marineris is basically the Mars Valley. And it's a huge canyon, way bigger than the Grand Canyon on Earth. OK. And it's a very striking scar. in the Martian surface. Clearly, Mars has had a whole lot of activity long ago, shaping and forming and reshaping this surface. And that's one of the bits of evidence of it. So that adds to the authenticity of what you got in your hands there. Very cool. I'm just going to let you talk while I sit here and play Black Van a White. Is that how that goes?

But I need a vowel though. I got a few of them right here. Oh yeah, there you go. I can tell you this that there's nothing more boring than people's luggage. Think about it. You're absolutely right. By the way, speaking of that. So now there's a piece of luggage that like that's good. I want one. There's some exciting looks. By the way, the cool thing about this, not only is it like it's Mars right here, which means that when it comes down. Can't argue with Mars. It comes down the chute, right? Okay, you are down conveyor belt Mars is coming down a conveyor belt You'll be able to

recognize yours and as a matter of fact you can turn to somebody and go get your eyes to Mars But don't take my luggage while you do Okay, very cool working on good stuff got a lot of stuff all guard. All right. All right, Jux. That's all we have into that was great. I love it So now you know and I never thought to analogize a gravity assist with a mugging. I'm glad I could be of service. Thank you, Chuck. Always, always. What's the brightest star in the nighttime sky? Are we talking my career here, Neil? Just saying.

I've been waiting for us to get around to this. How bright is your star, Chuck? Yeah, no, if it were me up in the sky, I would just be a dim spot in the Milky Way. It is funny that a fundamental word in all of Hollywood is derived from my profession. It is. Astrophysics. It is the most fundamental word. It is. Are you a star, meaning are you burning thermonuclear fuel in your core? Exactly, right. And the walk of fame, it's a star. It's a star. That's all astro. Yeah. Well, it's said that from everything I've ever heard. That's the North Star. That's what people say. Well, that's the North Star. You can always find your way home because it's the brightest star in the sky. Just follow the North Star. Yes. So in my experience, nine out of 10 people will say that. Yeah. OK. They'll see the first bright star after sunset and then assert that it's the North Star. So in 100% of those cases, they're looking at a planet. Well, that makes sense because a planet is right there.

Yeah, well. Planets right in front of our noses. Planets are right here, man. But for me, the funny part is they'll make a wish on the, you know, star bright, star bright, first star I see tonight. Right. And then they make a wish. And of course, the wishes don't come true because they're wishing on planets. That's why you never knew that. I want to explain that to any kid now. And then if you do not. If you're not wishing on the star, and because you know that they're planets, it means you know enough science not to be wishing on stars. That's pretty cool. It all balances out. And it's so funny that you say that. It just popped into my head what a culturally entrenched sentiment that is, wishing on a star.

I mean, there's songs about it. It transcends geography and culture. You have it in pretty much all kinds of writings wishing on a star. That's pretty wild. Yeah. And I'm honored that my field supplies this level of cultural referencing to all that we do. So the North Star is not the brightest star in

the night sky. Okay, I just want to make that clear. Okay. Now. So now here's the thing. Yeah. Cause what, what is the North Star? Like, what is the North Star? And why is it? So let's get why is it of any significance if it's not the brightest star? There you go. So we'll get to the brightest star in a minute. So the North Star is the star in the sky that's closest to where Earth's axis points on the sky. So you have a rotating Earth, and it's just sort of rotating. And the axis is just sort of sitting there as Earth spins around it. That axis is pointing to a spot in space, in the sky. So we say, is there a star near to that spot? Because then we'd always be pointing in that direction. So the star closest to that spot is the North Star.

And the North Star has a name. Did you know this? I want to guess because the only star that I hear hear the name of all the time is the dog star. Yeah. No, that's not. OK, well, there you go. I said it was a guess. It's not serious. It's not serious. The dog star. OK. Yeah. Yeah. The dog story is serious, serious of the dog star, not the North Star. So the name of this star is sensibly Polaris. Oh, for the pole or the axis, Polaris. Polaris. And in fact, that was the name I think of America's first nuclear submarine, Polaris, that could launch intercontinental ballistic missiles. We'd love to drop a star on you. So American. I know. That is so American.

Okay, we're about to drop a star on you. That produces energy to vaporize you just the way the centers of stars do. All right, that's the same thermonuclear mechanism is going on. So anyway, so it's got a name Polaris. Now the big dipper, okay, the lip of the big dipper, this part right here, okay, the lip. So there's a handle and then there's the cup and the lip is the front edge of that cup. Right. That points to the North Star. OK. OK. That's how you can always find the North Star. Because the Big Dipper is very obvious in the Northern Hemisphere. Absolutely. OK. So you feel and then it points and you go like four segments up and then boom. There it is. OK. So there are a couple of issues with this. First, the North Star is not exactly above the pole. OK. You can fit.

two full moon widths between it and the exact spot over the North Pole. Gotcha. All right, which means if you took a camera and a long exposure photo centered on the North Star, the North Star will not be pegged in the middle. You'll see it make a circle. Interesting. So it is not exactly above the

North Star. And the reason why you're seeing it make a circle is because you have the long exposure and the Earth is rotating where your camera is. Yes, exactly. And so, otherwise, if you just take snapshots, you just get the stars where they are. But if you take a long exposure, you get the stars blurring, basically. But they blur with the rotation of the Earth. Gotcha.

And so you'll see that it actually would trace a circle around the actual North Pole because people are saying, isn't it amazing that we have a star exactly over the North Pole? What are the odds? Well, it's not exactly. The odds would be zero. That's a bad bet. Yeah, it's not exactly over the noble. Not only that, the lip of the Big Dipper kind of misses the North Star. If you did an exact line, it's off. So there's a lot of fakery going on, not bad fakery, but just people want the sky to have more meaning than it actually does. And so you say things that are sort of partly true just because you feel better about what it is you're talking about. Okay, so the North Star is kind of like directionally, it's kind of like getting directions in the hood.

He's kind of over there. Yeah, yeah, man. So what you want to do? Are you going to keep going down there? You will keep going and going, right? Right? Then you're going to come to... So it's a car, right? But one of the wheels is up on a crate? Yeah, make it right there. Make it right there. You're going to keep going, right? Just keep going and going, right? That's right. Directions on the sky from the stars are exactly like that. So now let me tell you how... Bright the North Star is it's not the brightest, but let me tell you take a guess I'm gonna say it's a B student bright Okay, it's not how do you measure brightness in this in the night sky I really don't even know that but we have meters that do this This is science that you don't have to worry about that we got we got this All right And even if you didn't have meters, you could say, yeah, that's brighter than that. And that's dimmer than that. Yeah, we got this. Okay. We've had this even since before they were telescopes. Okay. So it is not in the top 10. Oh no. It is not in the top 20. Oh goodness. It is not in the top 30. Dad, North Star. What are you doing? It is not even.

in the top 40. Oh my god, the North Star is like the United States with respect to math and science. We say we're number one, but we're 47. The North Star is the 49th brightest star in the night sky. Oh my gosh. It is

completely uninteresting. It is easily missable. It does not call attention to yourself. That's why you need the lip of the Big Dipper to find it. To find it. Because you actually need a pointer because it really isn't remarkable at all. It is not remarkable. Not remarkable. Wow. It kind of reminds me of somebody I know. You sound like you should have your own TV show. Reminisances of everyone you grew up with. Stories. So yeah, so the North Star is just lame. I mean, it's just embarrassing.

And meanwhile, people think it's the brightest star in the night sky. And that's what's weird about it to me. Because they've never checked. And so this is the lore overriding people's curiosity. I wonder how Venus feels about that. Venus is the most mistaken object for the North Star because it's the brightest in the night sky, the morning sky. That's what I'm saying. Venus is sitting around like, ah, just like a man. Just like a man. Here I am, just as bright as I can be. Every single night, I come out. Is that what Venus sounds like? Look at me, and you know what they say? Oh my God, look at the North Star. Once again, a man just taking my credit for the hard work that I do. Okay. And no, there doesn't happen to be a star in the South Pole. It's just a big empty spot. There is a star closer to the South Pole than the North Star is to the North Pole, but it's even dimmer. And we don't call that the South Star?

You can, but it has a name, it's called Sigma Octans, and it's very unassuming and nobody cares. You know why nobody cares? Because its name is Sigma Octans. Don't nobody want to talk about Sigma Octans! Because there's a constellation called Octans, which is an octant, which is an early version of a sextant, and there's a lot of... navigational instruments among the constellations of the Southern Hemisphere. You know why? Because when Europeans got to the Southern Hemisphere and decided to map the stars, we had already begun the Industrial Revolution. And so they weren't thinking centaurs and Greek mythology, they were thinking, oh my gosh, I'm going to put some badass equipment here. So there's an architects table. There's a telescope, a microscope.

a sextant and an octant. Wow. It's all there. These are all constellations of the 88 in the night sky. Wow. So one last thing about the North Star. So you can picture this. If you are Santa Claus looking straight up, what star do you

see? Looking straight up, you see a dead spot. Well, yes, because we just learned that. Right. I thought it was a trick question. No, that's not true. Santa Claus will look up and see the North Star. The North Star, right. OK, so how many degrees up is that from the horizon? Yeah, from the horizon. Well, no, that should be nothing. He's already there. No, no. Degrees from the horizon to straight overhead for Santa Claus. Let me see, because he's at the top of the world. So the degrees on the horizon. No, forget it. It doesn't matter. Just yeah, Santa Claus is on the.

Quote, top of the world, sure. But now he wants to know how high up is the North Star for him above the horizon. So how many degrees is that? Oh, that's straight up. That's 90 degrees. 90 degrees. What is the latitude of the North Pole? 360 degrees. What is the latitude of the North Pole? 90 degrees. 90 degrees. Thank you. No, I said that because I wanted to see. This is what I love. This is what I love you being the educator that you are because I'll do that and I'm just waiting for the reaction because if it were me, I'd be like, man, what the hell is wrong with you? Hey, you do this. This is what you do. I gave you a second chance. You always do this. You're like.

I said my boy knows better than this. I'm going to give him a second chance. I'm not going to say a damn thing. He's going to fix his own damn mistake. But I love it. I love it. Go ahead. The North Star is 90 degrees up when you're at the North Pole, which is at a latitude of 90 degrees. That is not a coincidence. OK. If Santa Claus marches south, the North Star will get lower and lower and lower in the sky. Santa Claus gets to the equator. The North Star is on the horizon. The latitude of the equator is zero. The elevation of something on the horizon is zero. So the elevation in degrees of the North Star above your horizon is your latitude on Earth. Oh, because it follows all the way down. It follows all the way down. That's pretty interesting. Yes, very interesting. And so this was an important navigational tool. That makes sense. For everybody, right.

Right, and of course the underground railroad follow the North Star because if you just walk towards the North Star you'll eventually reach Santa Claus, but Ohio is good enough Who knew that you're that the North Star was such a poser that very much a poser Actually, it's not its fault. It's people want it

to be so special and it just isn't so I want to I have more to tell you about how unspecial it is. Oh, awesome. We're 49th. We're 49th. We're 49th. Yes, for those who missed it, the North Star is the 49th brightest star in the night sky. Right. And in the night sky, it's not saying, here I am. It is not doing that. All right, so, but wait, there's more. Earth spinning on its axis, and it's at its north pole, points upwards, close-ish to where the North Star is on the sky.

All right. What I didn't tell you is, I don't know, no one has them anymore. You have a play with tops as a kid. I did. That's how old I am. Of course. You play the top and it spins and then eventually begins to wobble. Right. Okay. Right. Okay. We have an official term for that. It's called precession. Precession. It's wobbling. Precession. That's an official term. So earth wobbles. Okay. Do we spin once? in how often? How long does it take to spin once? 24 hours, I guess. Thank you. Yeah. If you want to be precise, it's actually 23 hours and 56 minutes and four seconds. Wow. In case you... But that's another explainer. Well, you know, why do we think it's 24 hours when it's actually that? Okay. I will... That's a whole other... We'll save that. Okay. If you have permission. Okay. All right. So, we spin once in 24 hours and we wobble.

And the wobble takes 26,000 years. The wobble takes 26,000 years. Correct. So we've only really completed one wobble since caveman days, right? Wow. You know, 30,000 years ago. Right. So that's slow. And it took some very careful measurements and brilliant thinkers throughout the last thousand years, 2000 years to figure this out. that this is actually happening in the actual sky. All right, so now, because we're wobbling, it means the North Pole of Earth isn't always pointing towards the North Star. And we called it Polaris. But wait a minute, if it's no longer, if later on it's not going to point to Polaris, not only is it not the North Star, should we keep calling it Polaris?

Or just sort of grandfather in well, that's what I once was back in the day exactly So if you look at that circle The diameter of that circle is in degrees is 23 and a half degrees Okay Okay, because that's how tipped we are from vertebrae. All right Okay, that's earth. That's what gives us our seasons the fact they were tipped. So we tip that and then we process around that central

point. All right, so if you draw that circle on the sky, yes it will go through the North Star because that's what it is now, but any other stars that it comes near? One of the stars in the night sky I think you've heard of, it's called Vega. Vega is one of the brightest stars of the night sky. It's about 30 light years away. We think there's a planetary system forming around it, we see a disc of material that's not quite discrete planets, but it's kind of, it's working its way towards forming planets. So we know some stuff about Vega. In 14,000 years, the wobble will take the North Pole axis close enough to Vega for that to be our pole star. Gotcha. And in ancient Egypt, that was long enough ago.

You know, 4,000 5,000 years ago. That's an appreciable fraction of the 26,000 years. They had a different poll star than the North Star. Right. So in recorded history, civilizations have had different poll stars. And only right now is it what we're calling Polaris. Wow. Okay, that is nothing magical or permanent about it. Wow. It's like being in Manuto. Does anyone under 50 even know what that reference means? No, they have no idea what I'm talking about right now. But for those of you who don't know, Minuto was a boy band and they just kept swapping out members. One of which was Ricky Martin. Oh, that forgot about that. That's right. Yeah, so Minuto was not a set of names. Right. It was a concept. It was a concept.

Yes, and it's always had it. You just swapped out the members as they were age out. They'd be like 18. You're age out. That's what it was. Yeah. Sorry, Manuel. Your time is up. You got to go, bro. Wait, wait. You are now a man. You're much more into the man band. You can't be in the boy band anymore. So that's pretty wild. Yeah, so the North Star is a concept, and sometimes you won't have one. When it's sort of between other two stars, we're not gonna have a North Star. That's dope. I love it. That is the most interesting thing about the North Star is that the North Star, one, is not the North Star, and two, it ain't even, it's not even gonna be there. It's not gonna be there in a few thousand years. And it's not even in the top 40. And it's not even the top 40. You know?

That's pretty wild. All right. That is totally wild. Now, there's one other thing. It's the con man of all stars. Don't blame. No, don't blame the star itself. You fraud. I got one more thing. The North Star is a fraud. Go ahead.

I got one more thing. You ready? All right. All right. One of the consequences of the procession. Right. is that the sun is in a different part of the sky at the same time of the year. In other words, right now, pick a month, okay? Right. Let's pick October. Okay. And so October, astrologically, the sun is moving through Libra. Okay. Okay. Well, the procession of the earth, the wobble, over the 26,000 years will systematically put the sun in a different constellation in the month of October. Right. Okay. So the entire zodiac rotates through the calendar. Wow. In fact, when the zodiac was codified two and a half thousand years ago, wait a minute, that's like one 12th of 26,000.

Oh my gosh. So is October still the month for Libra? No! The sun was in Libra 2,000 years ago in October. All of that has shifted. And so, if you thought you were Libra, you're actually Scorpio right now. And that's when the sun gets all sexy. I didn't know you were fluent in this. So the Zodiac people, the astrologers, still describe the sun being in these various constellations as they were mapped out 2,200 years ago, between 2,000 and 2,400 years ago. And so if you're cancers, Scorpio, Gemini, you would expect the sun to be in that constellation the day you were born. 2,200 years ago. Oh man, that is so... It is not today. It's all shifted. And they don't really... They're not really forthcoming about that. No, because if they were, then all of astrology would be... Oh crap! That is... Oh man. Not only that, not only that, the zodiac... The sun in the zodiac actually passes through a 13th constellation. It's called Ophiuchus.

All right, God bless you. Ophiuchus. Ophiuchus. Wow, okay. In fact, the sun spends more time in Ophiuchus than it does in Scorpius. Okay. So, if you thought you were Scorpio, you were probably Ophiuchan, and all Scorpios and Ophiuchans are currently Librans. Wow. I just thought I'd put that out there. I just want to know where does that leave cancer, because that's what I am. Oh, your cancer? Yeah. I'm a cancer, so. So cancer is one of the lamest constellations ever constructed in the night sky. Damn. There's like five stars, and they have like a crab with claws, and it's a crustacean. It's like, what were you smoking? What opium was working in your system to say, oh, let's call that a crab. Man, you just, I can't believe this. It's like one.

I got a lousy constellation. Two, they may plus those stars are very bright. And they're not. OK, I got a lot. If you ask if you ask astrologers, they they're 12 prominent constellations in the sky. No, they're not. There's like three that kind of look like what they're supposed to. I got a lousy dim constellation. They made me into a stupid bottom feeder crab. And on top of that, I'm a festering malignancy. So yes, which they named after crabs. Yes. Oh, God. Yes. Oh, man. Cancer. Is it too late to be a Sagittarius? Where's cancer now? I don't know. I don't keep it. I've memorized that. I'll come back to you on another episode. OK. All right, we got to call it quits there. Just to round out everything you didn't need to know about the North Star and its brethren. Very cool. Star talk, explainers. Neil deGrasse Tyson. Keep looking up.